Lab 10 – External Flow over a Cylinder: Recorded Data

Grant Bellon

Environmental Conditions

Quantity	Value	Uncertainty
Barometric pressure, P_{atm}	999.7 mbar	± 0.1 mbar
Room temperature, T_{atm}	27.1 °C	± 1 °C

Water Density Sample

Quantity	Value	Uncertainty
Water volume, Vol_{water}	24 mL	± 0.5 mL
Water mass, m_{water}	24.2 g	± 0.05 g

Geometry

Quantity	Value	Uncertainty
Cylinder diameter, D	24.9 mm	± 0.1 mm
Cylinder length, $L_{cylinder}$	5 7/8 in	± 1/32 in
Water channel width, $w_{channel}$	6 1/4 in	± 1/32 in
Wetted height, h_{wetted}	5 1/8 in	± 1/32 in
Bearing to eyebolt, L_1	2 1/16 in	not recorded
Bearing to cylinder center, L_2	6 1/4 in	± 1/32 in

Drag Force Runs

Motor controller setpoint	Flowrate, Q	U_Q	Tension, T	U_T
10	52.7 GPM	± 0.5 GPM	0.146 N	± 0.01 N
20	106.2 GPM	± 0.5 GPM	1.02 N	± 0.01 N
30	162.8 GPM	± 2 GPM	2.86 N	± 0.1 N
40	208.2 GPM	± 3 GPM	3.68 N	± 0.1 N
50	223.5 GPM	± 5 GPM	3.98 N	± 0.1 N
60	247.8 GPM	± 7 GPM	4.73 N	± 0.2 N

Flow pattern sketches for each setpoint are on the original data sheets that follow.

Original Data Sheet, Page 1

...Fluids Laboratory

Laboratory Exercise 10 – External Flow Over a Cylinder

Experiment Data Sheet

Name: Grant Keller Date: _____

Names of Lab Partners (for reference):

Tim Straker	Peter Fernandez

Environmental Conditions:

Barometric Pressure, P_{atm}	Barometric Pressure Uncertainty, U_{P_atm}	Room Temperature, T_{atm}	Room Temperature Uncertainty, U_{T_atm}
997.7 mbar	± 0.1 mbar	23.1 °C	± 1 °C

Water Density:

Water Volume, Vol_{water}	Water Volume Uncertainty, U_{Vol_water}	Water Mass, m_{water}	Water Mass Uncertainty, U_{m_water}
24 mL	± 0.5 mL	24.2 g	± 0.5 g

Geometry:

Cylinder Diameter, D	Uncertainty of Cylinder Diameter, U_D	Cylinder Length, $L_{cylinder}$	Uncertainty of Cylinder Length, $U_{L_cylinder}$
24.8 mm	0.1 mm	5 7/8 in	1/32 in

Water channel width, $w_{channel}$	Uncertainty of water channel width, $U_{w_channel}$	Wetted height of water channel, h_{wetted}	Uncertainty of water channel wetted height, U_{h_wetted}
6 1/4 in	1/32 in	5 1/8 in	1/32 in

L_1 (see diagram)	Uncertainty of L_1	L_2 (see diagram)	Uncertainty of L_2
2 1/16 in		6 1/4 in	1/32 in

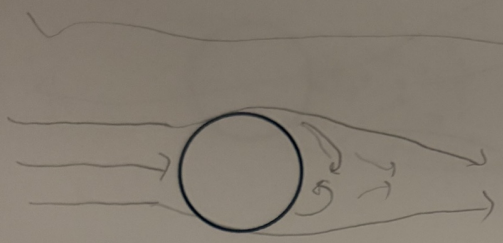
Original Data Sheet, Page 2

Drag Force and Flow Pattern Observation:

Motor Controller Setpoint = 10:

	Parameter	Uncertainty
Flowrate, *Q*	52.7 G/min	± 0.5 G/min
Tension Force, *T*	0.116 N	0.01 N

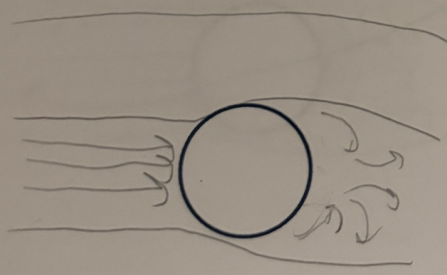
Sketch of Flow Pattern, with Approximate Flow Separation Angle:



Motor Controller Setpoint = 20:

	Parameter	Uncertainty
Flowrate, *Q*	108.2	± 0.5 G/min
Tension Force, *T*	1.02 N	± 0.01 N

Sketch of Flow Pattern, with Approximate Flow Separation Angle:

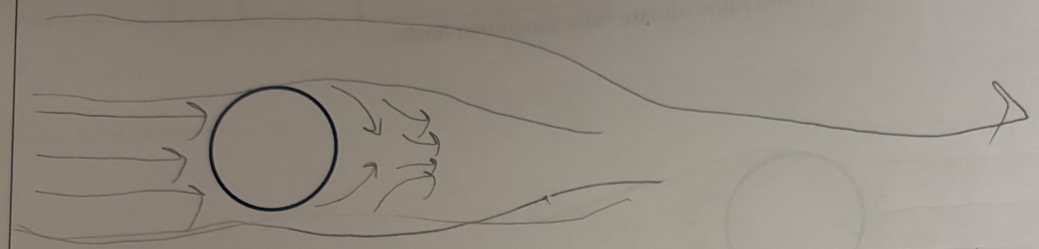


Original Data Sheet, Page 3

Motor Controller Setpoint = 30:

	Parameter	Uncertainty
Flowrate, *Q*	162,8	± 1,02 g/min
Tension Force, *T*	2,86	± 0,11 N

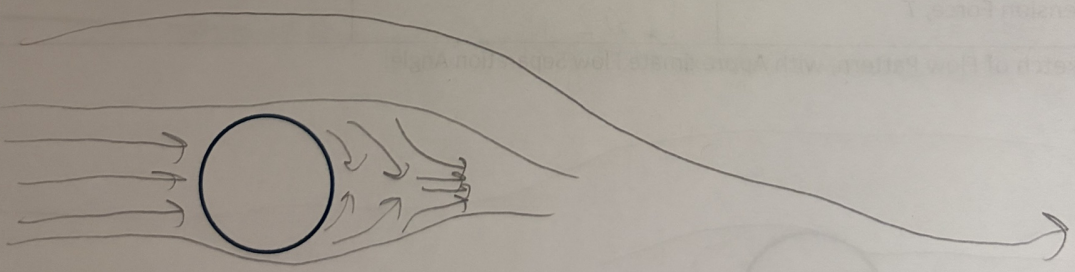
Sketch of Flow Pattern, with Approximate Flow Separation Angle:



Motor Controller Setpoint = 40:

	Parameter	Uncertainty
Flowrate, *Q*	209,2 g/min	± 3,9 g/min
Tension Force, *T*	3,68 N	± 0,11 N

Sketch of Flow Pattern, with Approximate Flow Separation Angle:

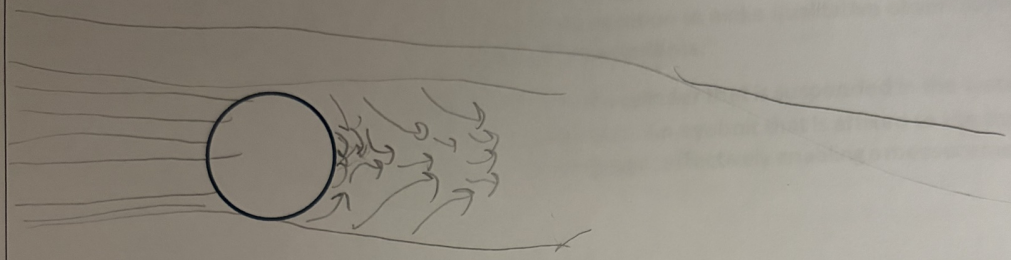


Original Data Sheet, Page 4

Motor Controller Setpoint = 50:

	Parameter	Uncertainty
Flowrate, *Q*	223,5 g/min	± 5,5 g/min
Tension Force, *T*	3,98 N	± 0,1 N

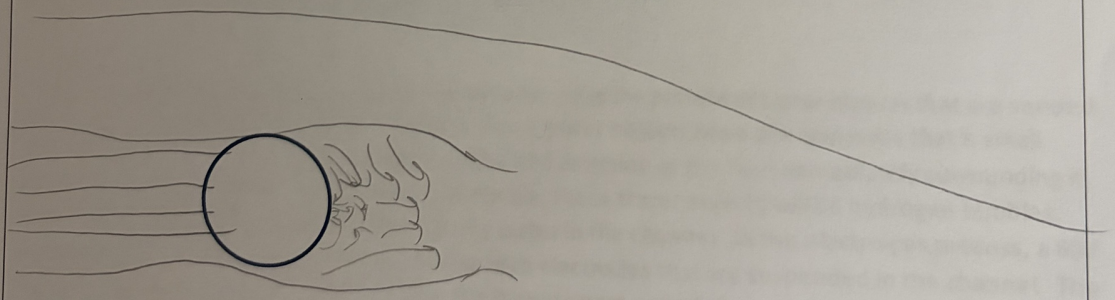
Sketch of Flow Pattern, with Approximate Flow Separation Angle:



Motor Controller Setpoint = 60:

	Parameter	Uncertainty
Flowrate, *Q*	247,5 g/min	± 7 g/min
Tension Force, *T*	4,17 g/min	± 0,2 N

Sketch of Flow Pattern, with Approximate Flow Separation Angle:



Instructor Signature: